

AF_XDP BUSYPOLL

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USECASE

- AF_XDP sockets bound to Isolated and dedicated queues
- HW Filters direct packets to these queues
  - Ethtool RSS Contexts
  - Application Device Queues
- Enable busy polling
  - napi_poll called from app context
  - Avoid Context Switches and Interrupts
- Application initiated polling of device queues

ndo_xsk_wakeup

- Called via poll() or send() system calls
  - When need_wakeup is set by the driver
- Triggers an interrupt to schedule NAPI
- Instead call napi_busy_loop(budget)
- Make busy poll budget configurable
  - The default value of 8 is too low for AF_XDP
  - sysctl net.core.busy_poll_budget

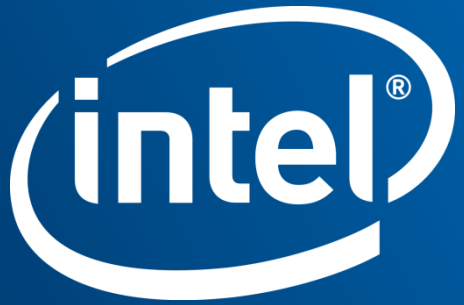
New NAPI STATE

- NAPI_STATE_NO_BUSYPOLL_STOP
- Set when AF_XDP Zerocopy Socket is bound to a device queue
- Cleared when the socket is detached from the queue
- Change busy_poll_stop()
  - Avoid calling napi_poll() when NO_BUSYPOLL_STOP is set
  - Do a napi_complete()

When to trigger busy poll

- Opportunistic busy polling
  - App thread is bound to the same core as queue vector's napi
- sysctl net.core.busy_poll is set
- Per socket SO_BUSYPOLL socket option is set

QUESTIONS/COMMENTS ?


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